

Entropy: $S = \log(\Omega)$ $\leftarrow$ # micro states

• In information theory:

→ Event A w/ prob. P_A

$$-\log(P_A) = S$$

Information Gained S

$$P = 1: S = 0$$

• This tells us that the event was determined and no sense of "randomness" or undetermined information was gained.

$$P = \frac{1}{\Omega}: -\log(P) = \log(\Omega) \quad \leftarrow \text{same as above}$$

$$A \cap B = \emptyset \quad \Rightarrow$$

$$-\log(P_A P_B) = -\log(P_A) - \log(P_B)$$

$$\sum_i P_i = 1$$

$-\log(P_i)$: information gained. Thus the expectation value is a sum weighted by this factor

$$\sum_i (-P_i \log(P_i)) = S_n : (\text{Shannon - Entropy})$$

$$(i) S \geq 0$$

$$S = 0 \Leftrightarrow$$

$$(ii) S \text{ is max if } P_i = \frac{1}{n}, \forall i$$

* Conditional Entropy: